JUMiner: An online literature mining tool for target identification and functional enrichment analysis

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ABSTRACT

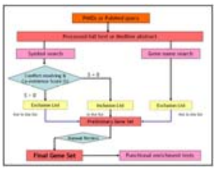
JUMiner is a web-based literature mining and functional analysis tool designed for mining of genes and protein products from MEDLINE abstracts and available full texts. It accepts a list of PubMed Unique Identifiers (PMIDs) as well as regular query strings in a PubMed Entrez search. JUMiner uses various rules by regular expression and a dictionary of gene symbols and descriptions compiled and expanded from multiple sources (HUGO and NCBI Gene (LocusLink)). The issue of name-conflict, in which same acronyms or abbreviations are often used for different genes, is taken care of by a scoring scheme based on the co-occurrence of acronyms and corresponding description terms. In order to further increase the accuracy, JUMiner accepts user-provided filters. JUMiner also supports functional enrichment analyses of targets (genes or proteins), GO (Gene Ontology) terms, MeSH (Medical Subject Headings) terms, pathways, and protein-protein-interaction networks against either another set of corpus or the whole documents in the JUMiner database. Performance was evaluated by using the BioCreAtIvE (Critical Assessment of Information Extraction systems in Biology) version 2 (Year 2006) Gene Normalization Task as a gold standard, which focuses on Human gene/protein correct identification. Out of 785 human gene/protein occurrences, JUMiner was able to identify 727 of them showing 93% recovery rate. 706 out of the recovered 727 were assigned to correct NCBI Entrez Gene IDs indicating 97% accuracy. Thus, JUMiner's literature mining performance coupled with functional enrichment analyses, will allow easy retrieval and summarization of rich biological information from corpuses of users' interest.

FEATURES & IMPLEMENTATION

A dictionary and rule based gene/protein mining tool, JUMiner implemented with MySQL, Perl, and CGI, has the following features.

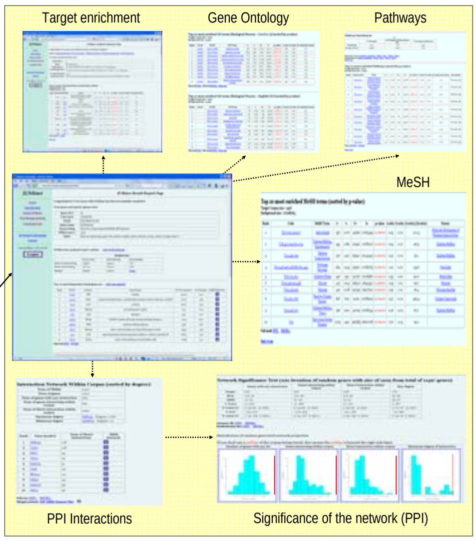
- **Primary dictionary** of gene names (124,823) and symbols (67,933)
  - Source: HUGO[1] and NCBI Gene DB[2] (LocusLink)
  - Expansion by rules to increase the coverage
- **Input**: A list of PMIDs or PubMed query
- **Text source**: Medline abstract and **full text html** if available
  - **Supported publishers** for full text retrieving: PubMedCentral, Nature, Science, Elsevier (Science Direct), Blackwell synergy, Informa world, Springer link, Ovid, Karger, and others
- **Scoring scheme** based on co-occurrence of symbols and names in text and MeSH RN
- **Web service**: http://jdrf.neurology.med.umich.edu/JUMiner/.
- **User-supplied additional filters** can be used to improve the accuracy
  - Ignore list (names), include list (symbols), and exclude list (symbols)
- **Post-mining functional enrichment analysis modules**
  - Mined targets can further be tested for their enrichments in **targets** (gene/protein), GO (Gene Ontology), MeSH (Medical Subject Headings), **pathways**, and **protein-protein interaction networks**.
  - Fisher's exact test is used.

WORKFLOW & SNAP SHOTS



Workflow

Query input:
"Reactive Oxygen Species"[MeSH] AND Diabetes



PERFORMANCE EVALUATION

A) Mining accuracy

BioCreAtIvE (Critical Assessment of Information Extraction systems in Biology)[3] version 2 (Year 2006) Gene Normalization Task. This comprises of 262 selected Medline abstracts with 785 occurrences of human genes.

	Count	Percentage
Recall	727	727/785 = 92.6%
Accuracy	706	706/727 = 97.1%

B) Name-conflict resolving power

NCBI GeneRIF (Gene Reference Into Function) providing human experts' functional annotation of genes described in Entrez Gene was used for the evaluation based on the assumption that conflicting symbols in GeneRIF descriptions have been confirmed by the human experts to be in proper context. Our in-house name-resolver (DavidNR) developed by Dr. David States has also been tested.

Date	03/05/08
Conflicting symbols (Total)	2,686
Conflicting symbols (with GeneRIF)	2,301
PMIDs	44,922
GeneRIF Sentences	77,236
Overlapping PMIDs	813
Symbols with overlapping PMIDs	249
Unique conflicting symbols (After removing overlapping PMIDs)	2,290

Symbol-mined-cases (original)	4,226,037
Symbols (total: Gene ID based)	1,812
Symbols (unique: Symbol based)	1,324

	PMIDs	TP	FP	TPR
JUMiner	13,894	12,603	1,291	0.907
DavidNR	11,306	10,306	946	0.916
23% more by JUMiner				

Better TPR			JUMiner			DavidNR		
(Out of 1,324 symbols)			Pos	TP	TPR	Pos	TP	TPR
JUMiner	427	32.3%	4657	4536	0.97	2753	2246	0.82
DavidNR	197	14.9%	3816	2768	0.73	3449	3089	0.90
SAME	682	51.5%	5389	5299	0.98	5104	5025	0.99
NO_Better	18	1.4%	32	0	0	0	0	0

C) Review-paper like gene-collection

JUMiner was tested how well it could collect genes like a review paper from a corpus of literatures. Li et al.'s review paper titled 'Epigenetics of prostate cancer'[4] was chosen for this test. The review paper has a summary table of genes related to epigenetics in prostate cancer. The following queries were used for JUMiner.

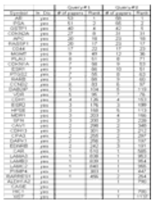
Epigenetic aberration	Gene symbol[1]
DNA hypermethylation	
Hormonal response	AR, ESR1, ESR2, RARB, RARRES1
Cell cycle control	CCND2, CDKN2A, CDKN1A, SFN
Tumor cell invasion/tumor architecture	APC, CAV1, CD44, CDH1, CDH13, LAMA3, LAMB3, LAMC2
Repair of DNA damage	GSTP1, MGMT
Signal transduction	DAB2IP, DAPK1, EDNRB, RASSF1
Inflammatory response	PTGS2
Others	ALDH1A2, HIC1, MDR1, PXMP4
DNA hypomethylation	CAGE, HPSE, PLAU, XIST
Histone hypoacetylation	CAR, CPA3, RARB, VDR
Histone methylation	DAB2IP, GSTP1, PSA

36 genes reported by Li et al.

Query#1: "Prostatic Neoplasms"[MeSH] AND "Epigenesis, Genetic"[MeSH]

Query#2: Prostate cancer epigenetics

	Papers	Recalled genes	%
Query#1	256	32	89%
Query#2	293	35	97%



Recalled genes by JUMiner

Enrichment results by JUMiner

GO Term (Molecular Functions)	genes	p-value	ratio
protein binding	661	4.1E-179	3.0
transferase activity	267	2.6E-54	2.9
nucleotide binding	202	2.4E-26	2.3
DNA binding	145	1.4E-24	2.6
ATP binding	142	2.7E-24	2.7
metal ion binding	266	2.4E-19	1.7
transcription coactivator activity	42	1.1E-18	6.3
transcription factor activity	104	1.3E-18	2.8
transcription repressor activity	35	1.5E-17	7.3
transcription activator activity	34	1.9E-14	5.8

GO Term (Biological Processes)	genes	p-value	ratio
cell cycle	131	3.8E-54	5.0
transcription	227	5.6E-40	2.7
regulation of apoptosis	56	1.3E-31	9.3
regulation of transcription, DNA-dependent	204	5.1E-31	2.5
protein amino acid phosphorylation	90	3.5E-27	4.2
regulation of progression through cell cycle	61	6.6E-27	6.4
apoptosis	75	1.0E-26	4.9
positive regulation of cell proliferation	51	1.7E-23	6.7
signal transduction	186	5.0E-23	2.2
negative regulation of cell proliferation	51	1.0E-22	6.4

Pathway Title	p-value	ratio
Prostate cancer	4.3E-40	14.9
Cell cycle	2.7E-33	11.3
MAPK signaling pathway	5.8E-30	6.2
Pancreatic cancer	3.6E-29	13.6
Focal adhesion	5.7E-29	7.4
Apoptosis	1.4E-28	12.4
Colorectal cancer	5.9E-28	11.9
Melanoma	8.2E-26	12.8
Small cell lung cancer	2.1E-24	10.9
Chronic myeloid leukemia	4.6E-24	11.7

SUMMARY

JUMiner is a convenient web-based platform for mining targets (genes and proteins) in full text literatures with functional enrichment analysis features. Powered with high recall rate (89~97%) and accuracy (91~97%) by name-conflict resolving algorithm, JUMiner can provide a unique opportunity of extracting knowledge from literature data.

REFERENCES

1) HUGO Gene Nomenclature Committee, http://www.genenames.org
2) Entrez Gene Database, http://www.ncbi.nlm.nih.gov/sites/entrez?db=gene
3) BioCreAtIvE challenge evaluation, http://biocreative.sourceforge.net
4) Li LC, Dahiya R: Epigenetics of prostate cancer. Front Biosci 2007

ACKNOWLEDGEMENTS

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